Andrew Schulz, PhD

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Research Website | Conservation Course | Humanitarian Course | Google Scholar

EDUCATION

PhD - Mechanical Engineering 2018-2022
Georgia Institute of Technology - Atlanta, GA

BS Mathematics 2013-2018
Oklahoma State University - Stillwater, OK

BS Mechanical Engineering 2013-2018
Oklahoma State University - Stillwater, OK

EXPERIENCE

Postdoctoral Researcher 2022-Present
Max Planck Institute for Intelligent Systems

- Research on Cheetah and Elephant Bio-inspired Robotics as well as conservation technology with Ardian Jusufi

Network Consultant 2022-Present
Engineering for One Planet

- Engineering for One Planet (EOP) is a global initiative to equip all engineers across all disciplines with the fundamental skills and principles of environmental and social sustainability.

Community Manager 2022-Present
WILDLABS Network

- WILDLABS is a global community of conservation technology educators, practitioners, developers, and change-makers. I am currently leading their education, teaching, and learning community.

Graduate Research Assistant 2018-2022
Georgia Institute of Technology

- Thesis: Mechanics, Materials, and Perception of the African Elephant Trunk with Applications to Conservation Technology, advised by Dr. David Hu

Executive Advisory Board 2021-Present
Alveus

- Advising a Wildlife conservation startup non-profit organization, Alveus Sanctuary based out of Austin, TX

Education Consultant 2020-2021
ConservationX Labs

- Developed the Conservation Future Toolkit on Technology for Environmental Impact, Online Course Module

AWARDS

- Best Paper Award - American Society of Engineering Education, Environmental Engineering Division
- Career, Research, and Innovation Development Conference - Provost Best Student Poster Award - 2022
- Student Recognition of Excellence in Teaching: 2021 CIOS Award
- Georgia Institute of Technology Presidential Fellowship - 2018, 2019, 2020, 2021
- Global RCE Network Award Winner - Building Effective ESD in Higher Education through Multi-Institution and Community Collaboration - 2021
- Vertically Integrated Project Instructor of the Year Award - 2021

GRANTS

5. Woodruff School New Faculty Seed Funding, "Biomechanics inspired conservation efforts through an undergrad led Machine Learning application", $25,000 (PI with Sabetta Matsumoto, 1/23-1/25).
4. Army Research Office URAP, "Creating Bio-Inspired Elephant Trunk Skin based on Tissue Mechanics Modeling", $9,000 (Co-PI with Dr. David Hu, 4/21-6/21).
3. European Hair Research Society Research Collaboration Grant, "African Elephant Sensing Capabilities through Proboscis Vibrissal Hairs", $2,500 (Co-PI with Dr. Claire Higgins, 4/19-12/19).
2. Smith gall-Watts Grant, "How Elephant Trunks Traverse an Array of Pegs", $14,000 (PI, 8/18 - 8/19).
1. Georgia Tech Research Institute, "Teaching STEM Majors about Sustainability, Public Health, and Wildlife Conservation", $25,000 (PI, 8/20-4/21).

TEACHING

Course Name	Semester	Role	Students	Effectiveness (X/5)
HumaniTech VIP	Fall 2022	IOR	26	4.99
HumaniTech VIP	Spring 2022	IOR	18	4.95
Tech4Wildlife VIP	Spring 2022	IOR	22	5
HumaniTech VIP	Fall 2021	IOR	25	4.93
Tech4Wildlife VIP	Fall 2021	IOR	15	5
HumaniTech VIP	Spring 2021	IOR	24	5

- VIP 2601/3601/4601/6601: **HumaniTech - VIP**
  Instructor of Record (IOR) for undergraduate and graduate students working on designing technology for public health concerns, including food deserts, nutrition education, and disaster relief. Average effectiveness as instructor 4.97/5.00 from 65 unique students.

- VIP 2601/3601/4601/6601: **GaTech4Wildlife - VIP**
  Instructor of Record (IOR) for undergraduate and graduate students working on designing technology for wildlife conservation, including reduction of invasive species, vaccination of urban wildlife, and zoo-collaborations for animal welfare. Average effectiveness as instructor 4.99/5.00 from 45 unique students.

INVITED TALKS

7. Berstein Center for Computational Neuroscience Berlin, Invited Talk to Department, 12/1/22
6. UMass Amherst, Neuroscience Graduate Student Speaker Series, 10/18/22
5. Northeast Ohio Medical University , Distinguished Lecture Series, 3/31/22
4. Boise State University, Biomechanics Lecture Series, 1/27/22
3. Animal Dynamics, Distinguished Lecture Series, 10/27/21
2. The Explorers Club, World Elephant Week, 8/10/21
1. The Smithsonian Institute, Invited Lecture, 6/16/21

LEADERSHIP AND SERVICE ACTIVITIES

- Postdoc Representative for SICB Division of Comparative Biomechanics - 2022-2024
- Organizing Chair for Southeastern Regional DCB-DVM SICB Conference 2021
- Co-Founder of GaTech4Wildlife Student Organization - 2021 - Present
- Founder of Woodruff School Graduate Mental Health Committee - 2019
- Organizing Chair for Zoo Biomechanics Day - April 2020, April 2023
- Journal of Experimental Biology Reviewer
- Elephant Managers Association Husbandry Committee Member - 2020 - Present